

COM3502/4502/6502 SPEECH PROCESSING

Lecture 08

The Fourier Transform

Fourier Transforms (Definition)

The Fourier Transform (FT) transforms continuous time signals to the continuous frequency domain.

$$\text{FT}\{x(t)\} = x(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

$$\text{IFT}\{x(j\omega)\} = x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(j\omega)e^{j\omega t} d\omega$$

The Discrete Time Fourier Transform (DTFT) transforms discrete (sampled) time signals to the continuous frequency domain.

$$\text{DTFT}\{x[k]\} = x(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} x[k]e^{-j\Omega k}$$

$$\text{IDTFT}\{x(e^{j\Omega})\} = x[k] = \int_{-\pi}^{\pi} x(e^{j\Omega})e^{j\Omega k} d\Omega$$

The Discrete Fourier Transform (DFT) transforms discrete time signals to the discrete frequency domain. (IDFT: Inverse DFT)

(FFT is no different transform but an

$$\text{DFT}\{x[k]\} = X[n] = \sum_{k=0}^{L_{DFT}-1} x[k]W_{L_{DFT}}^{kn} \quad \text{with } W_{L_{DFT}} = e^{j2\pi/L_{DFT}}$$

efficient algorithm for the DFT!)

$$\text{IDFT}\{X[n]\} = x[k] = \sum_{n=0}^{L_{DFT}-1} X[n]W_{L_{DFT}}^{-kn}$$

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Complex Numbers: A Reminder

$$z = x + jy$$

Imaginary axis

Real axis

unit circle

$y = \sin(\theta)$

$x = \cos(\theta)$

$z = e^{j\theta}$

θ

Magnitude $|z| = \sqrt{x^2 + y^2}$

Phase $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$e^{j\theta} = \cos(\theta) + j \sin(\theta)$

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Using the DFT

Signal:

N samples

Spectrum:

Result: N complex values

Frequency resolution = $\frac{1}{NT}$ (Hz)

Magnitude

Phase

frequency (Hz)

$f_s = \frac{1}{T}$

Symmetric about $\frac{1}{2T}$ Hz

Antisymmetric about $\frac{1}{2T}$ Hz

This is the explanation for Nyquist's theorem

N: DFT length

T: sampling interval

f_s : sampling frequency

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Implicit Periodicity in the DFT

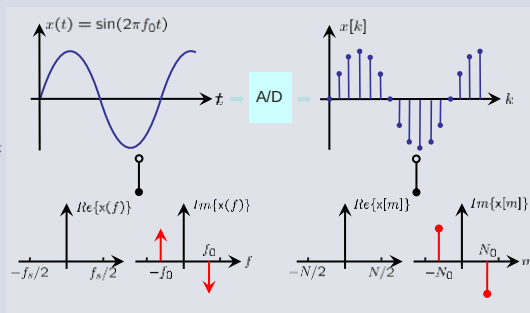
- The DFT computes the spectrum at N evenly spaced discrete frequencies
- The algorithm assumes periodicity outside the analysis frame (*with a period equal to the frame length*)
- This means that discontinuities will arise for ...
 - periodic signals with a *non-integer* number of cycles in the analysis frame
 - all aperiodic signals
 - all stochastic signals
- Such discontinuities give rise to unwanted spectral components

FT vs. DFT

continuous
time domain.

$$x(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

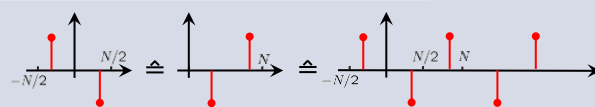
continuous
freq. domain.



discrete
time domain.

$$x[n] = \sum_{k=0}^{N-1} x[k]e^{j2\pi kn/N}$$

discrete
freq. domain.



In (e.g.) MATLAB / Numpy: first pos. freqs., then neg. frequencies

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Implicit Periodicity in the DFT

The diagram shows a signal waveform with an integer number of cycles within an analysis frame of N samples. The signal is assumed to be periodic by the DFT. The DFT is applied, resulting in a spectrum with a single peak at the desired frequency, indicating no spectral leakage.

Integer number of cycles

Analysis frame (N samples)

Signal as assumed by DFT.

Implicit periodicity

DFT

Spectrum of assumed signal. (as desired)

Amplitude

Frequency

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Implicit Periodicity in the DFT

The diagram shows a signal waveform with a noninteger number of cycles within an analysis frame of N samples. A discontinuity is present at the frame boundary. The DFT is applied, resulting in a spectrum with multiple peaks, indicating spectral leakage.

Noninteger number of cycles

Analysis frame (N samples)

Signal as assumed by DFT.

Discontinuity

Implicit periodicity

DFT

Spectrum of assumed signal. (has unwanted components)

Amplitude

Frequency

Spectral Leakage Effect
Undesirable components due to discontinuity

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Windowing

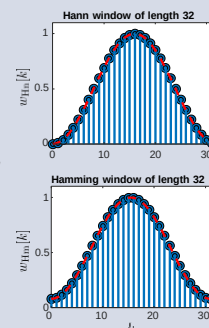
- The discontinuities arising from segmenting the signal into frames distorts the spectrum
- The distortion can be reduced by multiplying each signal frame with a 'window function'
- The most common window functions are

- the 'Hann (Hanning) Window' (raised cosine window, proposed by Julius von Hann)

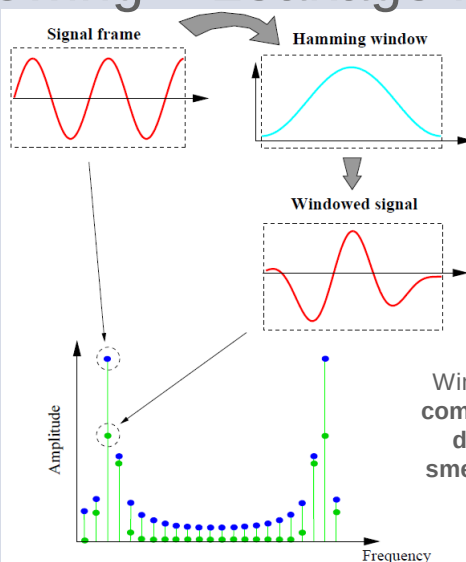
$$w_{\text{Hn}}[k] = 0.5 \cdot \left(1 - \cos\left(\frac{2 \cdot \pi \cdot k}{L}\right)\right)$$

- the 'Hamming Window' (proposed by Richard W. Hamming) ...

$$w_{\text{Hm}}[k] = 0.54 - 0.46 \cdot \cos\left(\frac{2 \cdot \pi \cdot k}{L-1}\right)$$

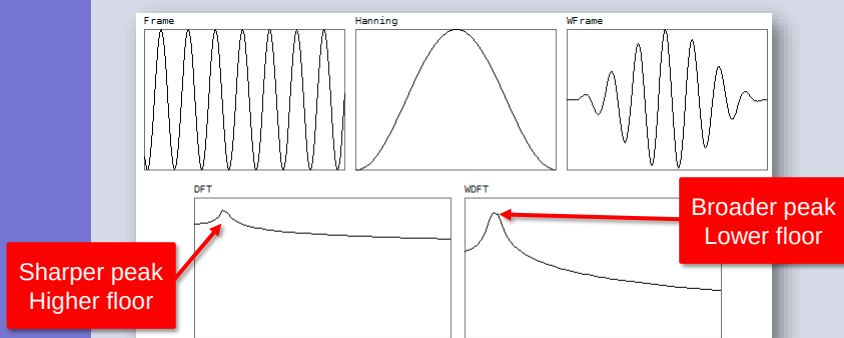


Windowing – Leakage Reduction



Windowing attenuates the components caused by the discontinuity, but also smears the spectral peaks

Windowing

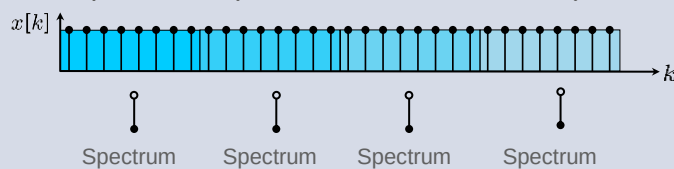


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Block-wise Frequency-Analysis



- If we analyze the spectral content on smaller sub-blocks, time information is preserved (with decreased resolution)



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Block-wise Frequency-Analysis

- If we analyze the spectral content on smaller sub-blocks, time information is preserved (with decreased resolution)

The diagram illustrates the process of block-wise frequency analysis. It starts with a signal $x[k]$ represented as a sequence of samples. A window function is applied to these samples, as shown by the overlaid bell-shaped curves. An inset graph titled 'Hanning window, 50% overlap' shows three such windows (blue, green, red) shifted by 10 samples each, demonstrating how overlapping windows preserve time information. Below, the signal is shown divided into blocks, each with its own spectrum calculated. Labels 'Spectrum' are placed under each block's spectrum.

Application of window function

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Block-wise Frequency-Analysis (Spectrogram)

This diagram shows how a spectrogram is created. It begins with a signal $x[k]$ and a window function inset (identical to the one on slide 30). The signal is processed into blocks, and their spectra are plotted against time. These individual spectra are then stacked to form a spectrogram. A 3D visualization shows the 'FREQUENCY' axis (vertical), the 'TIME' axis (horizontal), and the signal waveform (depth). A 2D spectrogram plot is also shown, with 'frequency in Hz' on the y-axis (0 to 4000) and 'time in seconds' on the x-axis (0 to 10). A color bar on the right indicates the magnitude of the frequency components, ranging from -50 to 20.

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Time vs. Frequency Resolution

Frequency resolution = $1/NT$ (Hz)

- Increasing the analysis frame N
 - decreases the spacing between the spectral components
 - reduces the ability to respond to changes in the signal
- Hence *large* N leads to **narrowband analysis**
 - good spectral resolution
 - poor time resolution

- Decreasing the analysis frame N
 - increases the spacing between the spectral components
 - increases the ability to respond to changes in the signal
- Hence *small* N leads to **wideband analysis**
 - good time resolution
 - poor spectral resolution

This is the time-frequency trade-off

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Time vs. Frequency Resolution

The figure displays two spectrograms of the sentence "The new brick factory fell over". The top spectrogram is labeled "WIDEBAND" and shows a dense, continuous pattern of energy across the frequency spectrum (0-5 kHz) over time (0-1.5 seconds). The bottom spectrogram is labeled "NARROWBAND" and shows a more sparse, discrete pattern of energy. Annotations on the narrowband spectrogram include: "Pitch rising" with an arrow pointing to a region where harmonics are moving fast; "Pitch falling" with an arrow pointing to a region where harmonics are moving slow; and "irregular very low pitch" with an arrow pointing to a region where the pitch is low and irregular. The narrowband spectrogram also shows "Section point (a)" and "Section point (b)" marked on the time axis.

Taken from: Holmes, J. N., & Holmes, W. (2002). *Speech Synthesis and Recognition*: Taylor & Francis.

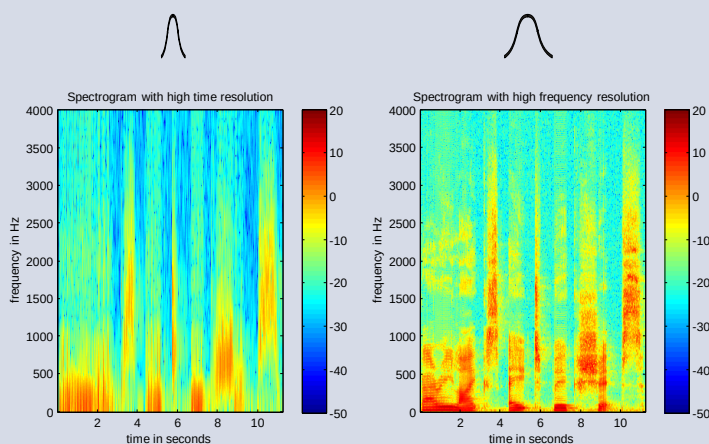
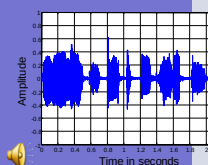
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Time vs. Frequency Resolution of the Spectrogram

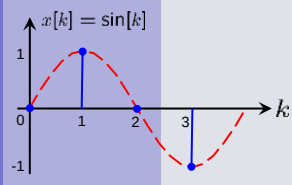


The Fast Fourier Transform (FFT)

- Implementation of the DFT requires the order of N^2 multiply-add operations
- By exploiting symmetry, it is possible to devise an algorithm that requires only $N \log_2 N$ multiply-add operations
 - e.g. for $N=2048$, the result is ~100x faster
- This more efficient algorithm is the ...
Fast Fourier Transform (FFT)
- The FFT requires that the window/frame should be a power of 2 in size
- This can be achieved by ...
 - choosing the appropriate analysis frame size, and/or
 - zero-padding a frame to the nearest power of 2

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DFT Example



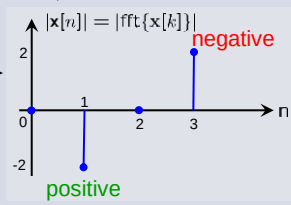
DFT:
$$x[n] = \sum_{k=0}^{N-1} x[k] e^{-jkn2\pi/N}$$

$$\begin{cases} x[0] = x[0] \cdot 1 + x[1] \cdot 1 + x[2] \cdot 1 + x[3] \cdot 1 = 0 \\ x[1] = x[0] \cdot 1 - x[1] \cdot j - x[2] \cdot 1 + x[3] \cdot j = -2j \\ x[2] = x[0] \cdot 1 - x[1] \cdot 1 + x[2] \cdot 1 - x[3] \cdot 1 = 0 \\ x[3] = x[0] \cdot 1 + x[1] \cdot j - x[2] \cdot 1 - x[3] \cdot j = 2j \end{cases}$$

Discrete Fourier transform matrix:

$$\mathbf{F} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \Rightarrow \mathbf{x}[n] = \mathbf{F} \cdot \mathbf{x}[k]$$

Matlab: `F = dftmtx(N)`



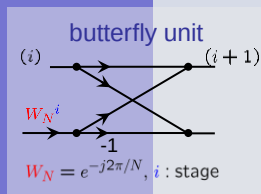
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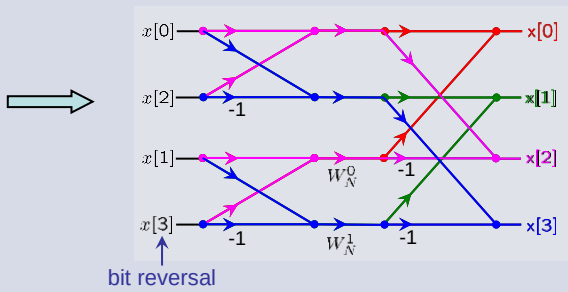
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Fast Fourier Transform (FFT)





bit reversal

FFT is no different transform but an efficient algorithm for the DFT!

$$\mathbf{F} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

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FFT (Bit Reversal)

index	binary	bitreverse	index
0	00	00	0
1	01	10	2
2	10	01	1
3	11	11	3

index	binary	bitreverse	index
0	000	000	0
1	001	100	4
2	010	010	2
3	011	110	6
4	100	001	1
5	101	101	5
6	110	011	3
7	111	111	7

Input samples (or output samples) have to be use in *bit-reverse* order!

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This slide set has covered ...

- Fourier Transform
- The discrete Fourier transform (*DFT*)
- Time versus frequency resolution
- Windowing
- The fast Fourier transform (*FFT*)

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Any Questions ?

*Raise during lecture or post in the
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Next time ...

Linear Filters

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